

Surface Dynamics of Noisy Viscoelastic Films by Adiabatic Approximation

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ABSTRACT: Surface dynamics is sometimes used to determine the rheological properties of soft materials. In typical data analyses, surface capillary waves are included without incorporating thermal noise. A phenomenological expression for the time-dependent power spectral density has been proposed to account for thermal noise and shown to agree well with experiment. In this paper, we investigate the surface dynamics of viscoelastic films with thermal noise by using an adiabatic approximation involving fast quasi-equilibrium elastic vibrations to derive the power spectral density. Our result justifies the use of the phenomenological expression.

$$\delta h(\mathbf{r}, t) = \underbrace{H(\mathbf{r}, t)}_{\text{Film height fluctuation}} + \underbrace{u(\mathbf{r}, t)}_{\text{fluctuations due to slow viscous flow}} + \underbrace{v(\mathbf{r}, t)}_{\text{fluctuations due to fast elastic vibrations}}$$

1. INTRODUCTION

The surfaces of fluids and soft materials are ever evolving due to the presence of thermally excited surface capillary waves.^{1–4} Experimental techniques, notably light scattering^{5,6} and X-ray photon correlation spectroscopy (XPCS),^{7–9} have been developed to measure the surface dynamics of soft materials and thereby their rheological properties. Recently, we developed a technique that measures the power spectral density (PSD) of the films at different times by using atomic force microscopy (AFM) and models the data based on the dynamics of surface capillary waves. With this technique, we have been able to determine the rheological properties of polystyrene films supported by silicon with thickness down to ~ 3 nm,^{10,11} where a significant effect of the free surface on the dynamics was found.^{10–13}

In light scattering or XPCS, the autocorrelation of the intensity fluctuations of the light or X-ray scattered from the sample surface is measured, and from the decay in the autocorrelation, the dynamics of the surface capillary waves is deduced. In our AFM technique, on the other hand, we measure the temporal evolution of the surface topography, as revealed by the PSD, of the films as they equilibrate from the (out-of-equilibrium) topography acquired upon preparation.^{14,15} If no external force is applied to the system as in our experiment, the evolution process is driven by thermal noise in addition to instability if present. We have previously found that merely assuming the deterministic dynamics of the surface capillary waves and neglecting the effect of thermal noise resulted in poor description of the experimental data.¹⁶ By drawing parallels between the dynamics of spinodal phase separation and thin liquid films, we constructed the time-dependent PSDs for the latter, $A_q^2(t)$, by borrowing the solution to the spinodal problem including thermal noise:¹⁶

$$A_q^2(t) = A_q^2(0) \exp(2\Gamma_q t) + \left(\frac{k_B T}{\gamma q^2 + G_0''} \right) [1 - \exp(2\Gamma_q t)] \quad (1)$$

where k_B is the Boltzmann constant, T is absolute temperature, γ and G_0 are, respectively, the surface tension and interfacial potential of the film, and Γ_q denotes the relaxation rate of the surface capillary mode with wavevector $\mathbf{q}$. Equation 1 has been found to produce excellent agreement with the experimental PSDs of not only low-molecular-weight, viscous polymer films,^{10,11} but also entangled, viscoelastic films^{12,13} as illustrated by Figure 1. The reason for the good agreement found for the latter is not obvious as the formalism of the spinodal phase separation problem and the viscoelastic film problem do not have a straightforward correspondence.

In the following, we shall derive an expression for the time-dependent PSD of viscoelastic films and thereby rationalize the applicability of eq 1. The derivation will be limited to cases where the amplitude of the surface fluctuations is much smaller while the wavelength of the surface capillary waves is much bigger than the average film height. The former allows linearization of the energy equation and the latter allows the pressure gradient in the film to be treated uniform in the out-of-plane direction. Experimentally, these limitations can be conveniently fulfilled by confining the study to the films with thickness $< \sim 100$ nm and PSDs taken in the initial stage of heating with $q < \sim 50 \mu\text{m}^{-1}$. To enable comparison with existing experiment,¹³ we shall focus on the films that exhibit the Maxwell liquid behavior, namely, ones that behave as an elastic solid at short times but a viscous liquid at long times. This is analogous to the dynamic behavior of entangled

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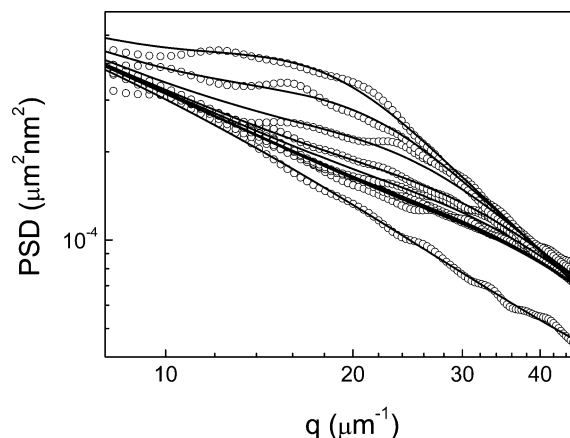


Figure 1. Power spectral density (PSD) of a 13-nm-thick polystyrene film ($M_w = 212$ kg/mol) coated on silicon with a 100-nm-thick SiOx upon annealing at 120 °C (symbols). The measurement times of the PSD are (from bottom to top): 0, 960, 3840, 7680, 30 720, 61 440, 184 320, 368 640, 737 280 s. The solid lines are the best fit to eq 1 where Γ_q assumes the expression in eq 31.

polymers that are initially in the rubbery state but crossover to the liquid state after the reptation time.^{12,13} To facilitate the calculation, an adiabatic approximation will be used wherein the surface capillary waves are first considered to be composed of wave components arising from elastic oscillations and viscous flow separately. Because the elastic displacements move at much shorter time scales than the viscous ones, we may treat the elastic displacements as following an equilibrium distribution dictated by the instantaneous amplitudes of the viscous displacements. This treatment has been found to reproduce eq 1 and an expression for the surface capillary dynamic rate derivable from solutions previously obtained for the time-dependent Navier–Stokes equation.^{4,17}

Organization of this paper is as follows. In section 2.1, we derive the equation governing the surface evolution of homogeneous supported films. In section 2.2, we discuss the solution to the evolution equation and incorporation of thermal noise. These discussions are intended to be pedagogical and lay background for the viscoelastic case. In section 2.3, we discuss the treatment of viscoelastic liquid films by using an adiabatic approximation. In section 3, we conclude this paper.

2. SURFACE DYNAMICS OF HOMOGENEOUS SUPPORTED FILMS

2.1. Evolution Equation for Homogeneous Supported Liquid Films in the Presence of Noise. Consider a homogeneous film with thickness $h(\mathbf{r})$ supported by a solid substrate, where $\mathbf{r}$ denotes the film's in-plane coordinate (see Figure 2). The unit-area surface energy of the film is $\gamma(\nabla h)^2/2$, where γ is the film's surface tension. Suppose $G(h)$ represents the unit-area interfacial potential; the total free energy of the film is given by

$$F = \int d^2\mathbf{r} \left[\frac{\gamma}{2} (\nabla h)^2 + G \right] \quad (2)$$

For thin apolar films, $G(h)$ is dominated by the van der Waals interactions and can be written as $G(h) = -A/(12\pi h^2)$, where A is the Hamaker function.¹⁸ Using this and eq 2, the local excess pressure is

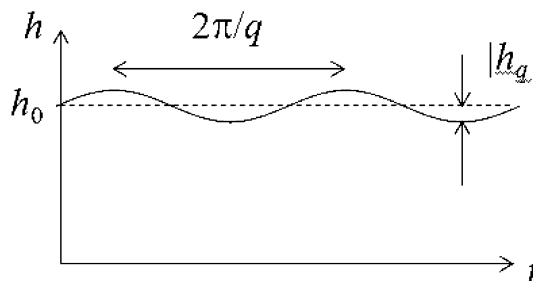


Figure 2. Diagram illustrating the experimental geometry. Note that the r -axis can lie in any direction in the film plane.

$$p(\mathbf{r}, t) = \frac{\delta F}{\delta h(\mathbf{r}, t)} = -\gamma \nabla^2 h(\mathbf{r}, t) + G'(h) \quad (3)$$

Assuming the no-slip boundary condition at the substrate interface and noting that the free surface is shear-stress free, this excess pressure induces a fluid current density of

$$\mathbf{j}(\mathbf{r}, t) = -M(h(\mathbf{r}, t)) \nabla p(\mathbf{r}, t) + \mathbf{s}(\mathbf{r}, t) \quad (4)$$

where

$$M(h) = \frac{h^3}{3\eta} \quad (5)$$

is the mobility of the film (see Appendix A for its derivation) and η the effective viscosity.¹² The last term in eq 4 denotes the thermally induced noise in the current density, which has zero mean value and correlation given by

$$\langle \mathbf{s}_k(\mathbf{r}, t) \mathbf{s}_l(\mathbf{r}', t') \rangle = 2D \delta_{kl} \delta^2(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (6)$$

where D is the noise magnitude; k, l are the coordinate indices, δ_{kl} is a Kronecker delta function, and $\delta^2(\mathbf{r} - \mathbf{r}')$ and $\delta(t - t')$ are Dirac delta functions. The evolution equation of the film profile is obtainable by writing the continuity equation

$$\frac{\partial h(\mathbf{r}, t)}{\partial t} = -\nabla \cdot \mathbf{j} = \nabla \cdot [M(h(\mathbf{r}, t)) \nabla p(\mathbf{r}, t)] + \xi(\mathbf{r}, t) \quad (7)$$

where $\xi(\mathbf{r}, t) = -\nabla \cdot \mathbf{s}(\mathbf{r}, t)$ and has also a zero mean value and correlation

$$\begin{aligned} \langle \xi(\mathbf{r}, t) \xi(\mathbf{r}', t') \rangle &= \langle \nabla \cdot \mathbf{s}(\mathbf{r}, t) \nabla' \cdot \mathbf{s}(\mathbf{r}', t') \rangle \\ &= \partial_i \partial'_j \langle \mathbf{s}_k(\mathbf{r}, t) \mathbf{s}_l(\mathbf{r}', t') \rangle \end{aligned} \quad (8a)$$

$$= 2D \partial_i \partial'_j \delta^2(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (8b)$$

$$= -2D \partial_i \partial'_j \delta^2(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (8c)$$

$$= -2D \nabla^2 \delta^2(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (8d)$$

In eqs 8a–8c, a summation over the repeated indices is assumed. In addition, use has been made of eq 6 in going from eq 8a to 8b.

2.1.1. Linearization and Fourier Transformation. Next, we linearize eqs 2 and 7 about $h = h_0$, the average thickness of the film, and obtain

$$F = \int d^2\mathbf{r} \left[\frac{\gamma}{2} (\nabla h)^2 + \frac{1}{2} G''_0 (h - h_0)^2 \right] \quad (9)$$

and

$$\frac{\partial h}{\partial t} = -M_0 \gamma \nabla^4 h + M_0 G_0'' \nabla^2 h + \xi \quad (10)$$

where $M_0 = h_0^3/(3\eta)$, $G_0'' = -A/(2\pi h_0^4)$ and a constant term in eq 9, which does not contribute to the dynamics, is dropped. Upon Fourier transformation, eqs 9 and 10 become

$$F = \sum_{\mathbf{q}} \left[\frac{\gamma}{2} \mathbf{q}^2 |h_{\mathbf{q}}|^2 + \frac{1}{2} G_0'' |h_{\mathbf{q}}|^2 \right] \quad (11)$$

$$\frac{dh_{\mathbf{q}}}{dt} = -M_0 \gamma \mathbf{q}^4 h_{\mathbf{q}} - M_0 G_0'' \mathbf{q}^2 h_{\mathbf{q}} + \xi_{\mathbf{q}} = \Gamma_{\mathbf{q}} h_{\mathbf{q}} + \xi_{\mathbf{q}} \quad (12)$$

where

$$\Gamma_{\mathbf{q}} = -M_0 \mathbf{q}^2 [\gamma \mathbf{q}^2 + G_0''] \quad (13)$$

It is noteworthy that, had the noise term $\xi_{\mathbf{q}}$ been absent, eq 12 would be identical to the classic result derived previously.^{19,20}

2.2. Solution to the Evolution Equation and Their Power Spectral Density. It is straightforward to show that the solution to the evolution eq 12 is

$$h_{\mathbf{q}}(t) = e^{\Gamma_{\mathbf{q}} t} \left[h_{\mathbf{q}}(0) + \int_0^t d\tau_1 e^{-\Gamma_{\mathbf{q}} \tau_1} \xi_{\mathbf{q}}(\tau_1) \right] \quad (14)$$

Multiplying eq 14 with the complex conjugate $h_{\mathbf{q}}^*(t)$ and averaging, we get an expression for the power spectral density

$$\begin{aligned} A_{\mathbf{q}}^2(t) &= \left\langle |h_{\mathbf{q}}(t)|^2 \right\rangle \\ &= e^{2\Gamma_{\mathbf{q}} t} \left[A_{\mathbf{q}}^2(0) + \int_0^t d\tau_1 \int_0^t d\tau_2 e^{-\Gamma_{\mathbf{q}}(\tau_1 + \tau_2)} \left\langle \xi_{\mathbf{q}}^*(\tau_1) \xi_{\mathbf{q}}(\tau_2) \right\rangle \right] \end{aligned} \quad (15)$$

Applying eq B.6 and noting that $\xi_{\mathbf{q}}^* = \xi_{-\mathbf{q}}$ we get

$$\begin{aligned} A_{\mathbf{q}}^2(t) &= e^{2\Gamma_{\mathbf{q}} t} \left[A_{\mathbf{q}}^2(0) + 2D\mathbf{q}^2 \int_0^t d\tau_1 e^{-2\Gamma_{\mathbf{q}} \tau_1} \right] \\ &= e^{2\Gamma_{\mathbf{q}} t} \left[A_{\mathbf{q}}^2(0) + \frac{D\mathbf{q}^2}{\Gamma_{\mathbf{q}}} (1 - e^{-2\Gamma_{\mathbf{q}} t}) \right] \end{aligned} \quad (16)$$

To gain insight about the physical meaning of $D\mathbf{q}^2/\Gamma_{\mathbf{q}}$ we consider the stable modes where $\Gamma_{\mathbf{q}} < 0$. As $t \rightarrow \infty$ where the system reaches equilibrium, eq 16 becomes

$$A_{\mathbf{q}}^2(\infty) = \frac{D\mathbf{q}^2}{-\Gamma_{\mathbf{q}}} \quad (17)$$

which represents the asymptotic value of the PSD at long times. We may alternatively derive $A_{\mathbf{q}}^2(\infty)$ by apply the equipartition law to eq 11 and obtain

$$A_{\mathbf{q}}^2(\infty) = \frac{k_B T}{\gamma \mathbf{q}^2 + G_0''} \quad (18)$$

where k_B is the Boltzmann constant and T is the absolute temperature. By comparing eqs 17 and 18 and using $M_0 = h_0^3/(3\eta)$ (see just above eq 11) and eq 13, one finds that

$$D = M_0 k_B T = \frac{h_0^3 k_B T}{3\eta} \quad (19)$$

Finally, by combining eqs 16 and 17, we obtain

$$A_{\mathbf{q}}^2(t) = A_{\mathbf{q}}^2(0) \exp(2\Gamma_{\mathbf{q}} t) + \left(\frac{k_B T}{\gamma \mathbf{q}^2 + G_0''} \right) [1 - \exp(2\Gamma_{\mathbf{q}} t)] \quad (20)$$

which is identical to eq 1. The first term is the usual normal mode solution. The second term is traceable to the thermal noise term introduced to the current density in eq 4. Physically, it depicts the system approaching the equilibrium value $A_{\mathbf{q}}^2(\infty)$ at a rate equal to the relaxation rate, $-\Gamma_{\mathbf{q}}$ of the stable normal modes.

2.3. Deriving the Power Spectral Density of Viscoelastic Films by Using Adiabatic Approximation. As mentioned in the Introduction, we shall be focused on viscoelastic films that behave as Maxwell liquids, namely, viscoelastic materials that have a single relaxation time separating the short-term elastic behavior (with elastic modulus μ_0) from the long-term viscous behavior (with viscosity η). One may find parallels of Maxwell liquids in mechanical systems comprising a series connection between an elastic spring and a plunger (Figure 3) and entangled polymers in the rubbery state

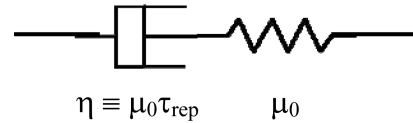


Figure 3. Diagram illustrating the mechanical model of a Maxwell liquid.

that enter the terminal flow regime after the reptation time, τ_{rep} . The dynamic behavior of a Maxwell liquid can be described by a frequency-dependent viscosity of the form^{4,17}

$$\eta(\omega) = \frac{\eta}{1 - i\omega\eta/\mu_0} \quad (21)$$

where $i = \sqrt{-1}$. By eq 21, the crossover between the elastic and viscous behavior occurs at $t = \eta/\mu_0$, which is thus identifiable with the relaxation time mentioned above. It should be mentioned that in the case of entangled polymers the rubbery state is preceded by the transition from the glassy state to the rubbery state that eq 21 does not model. Therefore, eq 21 applies only beyond this transition.

We suppose that the evolution in the film surface is caused by a superposition of capillary waves $H(\mathbf{r}, t)$ and $u(\mathbf{r}, t)$ due to viscous flow and elastic vibrations, respectively, and write the fluctuations in the film height as

$$\delta h(\mathbf{r}, t) = H(\mathbf{r}, t) + u(\mathbf{r}, t) \quad (22)$$

According to eq 11, this profile has an energy term, $\sum_{\mathbf{q}} (\gamma \mathbf{q}^2 + G_0'') |h_{\mathbf{q}}|^2$, attributable to the surface tension and interfacial potential. For viscoelastic films, there should be an energy associated with the elastic deformation as well. Taking these into account, the total energy of the film in the long-wavelength limit ($qh_0 \ll 1$) is

$$F\{H, u\} = \frac{1}{2} \sum_{\mathbf{q}} [\Phi_{\mathbf{q}} |H_{\mathbf{q}} + u_{\mathbf{q}}|^2 + \Psi_{\mathbf{q}} |u_{\mathbf{q}}|^2] \quad (23)$$

where $\Phi_{\mathbf{q}} \equiv \gamma \mathbf{q}^2 + G_0''$, $\Psi_{\mathbf{q}} \equiv 3\mu_0/(h_0^3 \mathbf{q}^2)$,²¹ and $u_{\mathbf{q}}$ and $H_{\mathbf{q}}$ denote the Fourier transforms of $u(\mathbf{r})$ and $H(\mathbf{r})$, respectively. With simple algebra, eq 23 can be rewritten as

$$F\{H, u\} = \frac{1}{2} \sum_{\mathbf{q}} \left\{ \Phi'_{\mathbf{q}} |H_{\mathbf{q}}|^2 + (\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}) \left| u_{\mathbf{q}} + \frac{\Phi_{\mathbf{q}}}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} H_{\mathbf{q}} \right|^2 \right\} \quad (24)$$

where

$$\begin{aligned} \Phi'_{\mathbf{q}} &= \frac{\Phi_{\mathbf{q}} \Psi_{\mathbf{q}}}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} \\ &= \left[\frac{1}{\Phi_{\mathbf{q}}} + \frac{1}{\Psi_{\mathbf{q}}} \right]^{-1} \\ &= \left[(\gamma \mathbf{q}^2 + G_0'')^{-1} + \left(\frac{3\mu_0}{h_0^3 \mathbf{q}^2} \right)^{-1} \right]^{-1} \end{aligned} \quad (25)$$

and $u_{\mathbf{q}}$ and $H_{\mathbf{q}}$ denote the Fourier transformation of $u(\mathbf{r})$ and $H(\mathbf{r})$, respectively. Next, we apply the adiabatic approximation where the fast-varying $u_{\mathbf{q}}$ is assumed to follow a quasi-equilibrium distribution dictated by the instantaneous value of the slow-varying $H_{\mathbf{q}}$. With this, one deduces that

$$u_{\mathbf{q}} = \langle u_{\mathbf{q}} \rangle + \delta u_{\mathbf{q}} \quad (26)$$

where

$$\langle u_{\mathbf{q}} \rangle = -\frac{\Phi_{\mathbf{q}}}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} H_{\mathbf{q}} \quad (27)$$

and $\delta u_{\mathbf{q}}$ follows a Gaussian distribution with zero mean value and variance equal to

$$\langle |\delta u_{\mathbf{q}}|^2 \rangle = \frac{k_B T}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} = k_B T \left[\gamma \mathbf{q}^2 + G_0'' + \frac{3\mu_0}{h_0^3 \mathbf{q}^2} \right] \quad (28)$$

Here, we further assume that $\gamma \mathbf{q}^2 + G_0'' + 3\mu_0/(h_0^3 \mathbf{q}^2) > 0$ for all values of $\mathbf{q}$. This is valid for sufficiently thick films so that G_0'' is not too negative. Otherwise, the film surface is unstable with respect to elastic deformation alone and one has to go beyond the linear stability theory adopted here. The free energy in eq 24, after averaging out the fast elastic degrees of freedom $u_{\mathbf{q}}$ is

$$F\{H\} = \frac{1}{2} \sum_{\mathbf{q}} \{ \Phi'_{\mathbf{q}} |H_{\mathbf{q}}|^2 + k_B T \} \quad (29)$$

The constant term $k_B T$ does not contribute to the flow and need not be considered. Equation 29 is then analogous to eq 11 and the dynamics of $H_{\mathbf{q}}$ is analogous to that of $h_{\mathbf{q}}$ with $\Phi_{\mathbf{q}}$ replaced by $\Phi'_{\mathbf{q}}$. One may thus write the analogous solution to $\langle |H_{\mathbf{q}}(t)|^2 \rangle$ based on eq 16

$$\langle |H_{\mathbf{q}}(t)|^2 \rangle = e^{2\Gamma'_{\mathbf{q}} t} \left[\langle |H_{\mathbf{q}}(0)|^2 \rangle + \frac{D \mathbf{q}^2}{\Gamma'_{\mathbf{q}}} \right] - \frac{D \mathbf{q}^2}{\Gamma'_{\mathbf{q}}} \quad (30)$$

where, similar to eq 13,

$$\Gamma'_{\mathbf{q}} = -M_0 \mathbf{q}^2 \Phi'_{\mathbf{q}} = -M_0 \mathbf{q}^2 \left[(\gamma \mathbf{q}^2 + G_0'')^{-1} + \left(\frac{3\mu_0}{h_0^3 \mathbf{q}^2} \right)^{-1} \right]^{-1} \quad (31)$$

It can be shown that eq 31, depicting the dynamic rate of surface capillary waves in the adiabatic approximation, is the same as that of the slow relaxation mode derivable by using the approach of Safran et al.¹⁷ for Maxwell liquids with dynamic behavior as depicted in eq 21. It is noteworthy that the dynamic rates $\Gamma_{\mathbf{q}}$ and $\Gamma'_{\mathbf{q}}$ from eqs 13 and 31 always have the same sign under the conditions specified above, implying that the film surface stability is not affected by viscoelasticity. This is expected since viscoelasticity only delays but does not prevent viscous flow.

Again, by requiring that the energy in eq 29 for the stable modes follows the Boltzmann distribution at thermal equilibrium, we obtain $\langle |H_{\mathbf{q}}(\infty)|^2 \rangle = k_B T / \Phi'_{\mathbf{q}}$, which upon comparison with eq 30 gives $D = M_0 k_B T$, the same as in the viscous case.

Next, we combine eqs 22, 26, and 27 and get

$$\begin{aligned} h_{\mathbf{q}} &= H_{\mathbf{q}} + \langle u_{\mathbf{q}} \rangle + \delta u_{\mathbf{q}} \\ &= \frac{\Psi_{\mathbf{q}}}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} H_{\mathbf{q}} + \delta u_{\mathbf{q}} \\ &= (\Phi'_{\mathbf{q}} / \Phi_{\mathbf{q}}) H_{\mathbf{q}} + \delta u_{\mathbf{q}} \end{aligned} \quad (32)$$

Since $H_{\mathbf{q}}$ and $\delta u_{\mathbf{q}}$ are uncorrelated and have zero mean value, we deduce from eq 32 that

$$\begin{aligned} A_{\mathbf{q}}^2(t) &= \langle |h_{\mathbf{q}}(t)|^2 \rangle \\ &= \left(\frac{\Psi_{\mathbf{q}}}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} \right)^2 \langle |H_{\mathbf{q}}|^2 \rangle + \langle |\delta u_{\mathbf{q}}|^2 \rangle \end{aligned} \quad (33)$$

By using the expressions of $\langle |\delta u_{\mathbf{q}}|^2 \rangle$ and $\langle |H_{\mathbf{q}}(t)|^2 \rangle$ in eqs 28 and 30, respectively, we obtain

$$\begin{aligned} A_{\mathbf{q}}^2(t) &= \left(\frac{\Psi_{\mathbf{q}}}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} \right)^2 \left\{ e^{2\Gamma'_{\mathbf{q}} t} \left[\langle |H_{\mathbf{q}}(0)|^2 \rangle + \frac{D \mathbf{q}^2}{\Gamma'_{\mathbf{q}}} \right] \right. \\ &\quad \left. - \frac{D \mathbf{q}^2}{\Gamma'_{\mathbf{q}}} \right\} + \frac{k_B T}{\Phi_{\mathbf{q}} + \Psi_{\mathbf{q}}} \end{aligned} \quad (34)$$

It should be remarked that under the presumed quasi-equilibrium among the elastic vibrations and by eq 28, $A_{\mathbf{q}}^2(0) \geq k_B T / (\gamma \mathbf{q}^2 + G_0'' + 3\mu_0/(h_0^3 \mathbf{q}^2))$ with the equality occurring at $\langle |H_{\mathbf{q}}(0)|^2 \rangle = 0$. By using the definitions of $\Psi_{\mathbf{q}}$, $\Phi_{\mathbf{q}}$ (see just above eq 24) and $\Gamma'_{\mathbf{q}}$ (eq 31), and with some algebra, one can show that eq 34 is identical to eq 20 with $\Gamma_{\mathbf{q}}$ replaced by $\Gamma'_{\mathbf{q}}$, or

$$\begin{aligned} A_{\mathbf{q}}^2(t) &= A_{\mathbf{q}}^2(0) \exp(2\Gamma'_{\mathbf{q}} t) + \left(\frac{k_B T}{\gamma \mathbf{q}^2 + G_0''} \right) \\ &\quad [1 - \exp(2\Gamma'_{\mathbf{q}} t)] \end{aligned} \quad (35)$$

The present finding that eq 35 has the same form as eq 20 or 1 is attributable to the same equilibrium power spectral density,

namely, $A_q^2(\infty) = k_B T / (\gamma q^2 + G_0'')$. The reason for the same $A_q^2(\infty)$ may not be immediately perceivable as the free energy of the former has an additional elastic energy term (see eqs 11 and 23). Mathematically, one can show that, upon application of the equipartition law, the elastic energy term, with wave amplitude u_q being independent of the wave amplitude h_q ($\equiv H_q + u_q$) of the first energy term, does not contribute to the equilibrium distribution of h_q . Physically, elastic motions are succeeded by viscous flow in Maxwell liquids. So, the surface structure brought about by elastic motions eventually gets washed out by viscous flow and does not contribute to the final equilibrium structure.

3. CONCLUSION

We have derived expressions for the time-dependent power spectral density of homogeneous supported films with non-negligible thermal noise. Results obtained for both viscous liquid films and viscoelastic Maxwell liquid films are presented. After adopting an adiabatic approximation for viscoelastic films, the evolution of the surface wave amplitude follows an exponential time dependence similar to the viscous case, except for the material-specific dynamic rate.

■ APPENDIX A. DERIVATION OF THE MOBILITY M OF THIN FLUID FILMS

The Navier–Stokes equation for incompressible fluid reads²²

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \frac{\eta}{\rho} \Delta \mathbf{v} \quad (\text{A.1})$$

where $\mathbf{v}(\mathbf{r})$ is the velocity of the fluid at position $\mathbf{r}$ and ρ is the fluid density. Consider steady-state, open-channel flow on an infinite plane, with a uniform fluid depth of h_0 , due to a uniform pressure gradient ∇p parallel to the plane. Without loss of generality, we suppose that the plane lies in the x – y plane and ∇p points in the x direction. With these, $\mathbf{v} = v(z)\mathbf{x}$ and $p = p(x)$, which make $(\mathbf{v} \cdot \nabla) \mathbf{v} = 0$ and $\partial \mathbf{v} / \partial t = 0$, and eq A.1 becomes

$$\frac{d^2 v}{dz^2} = \frac{1}{\eta} \frac{dp}{dx} \quad (\text{A.2})$$

Assuming the no-slip boundary condition, i.e., $v(0) = 0$, at the bottom ($z = 0$) and stress-free condition, i.e., $dv/dz = 0$, at the upper free surface ($z = h_0$), eq A.2 can be solved to give

$$v(z) = \frac{h_0^2 \nabla p}{2\eta} \left[\left(\frac{z}{h_0} \right)^2 - 2 \left(\frac{z}{h_0} \right) \right] \quad (\text{A.3})$$

The mobility, M , is defined to be the fluid current density per unit pressure gradient that causes it. By using eq A.3, we get

$$M = \frac{\frac{1}{h_0} \int_0^{h_0} v(z) dz}{-\nabla p} = \frac{h_0^3}{3\eta} \quad (\text{A.4})$$

The negative sign in eq A.4 arises because the fluid current flows from the high pressure side to the low pressure side and so points in the direction of $-\nabla p$.

■ APPENDIX B. NOISE CORRELATION IN THE MOMENTUM SPACE

We write the spatial Fourier transformation of the noise term, $\xi(\mathbf{r}, t)$, in eq 7 as

$$\xi(\mathbf{r}, t) = \Omega^{-1/2} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}} \xi_{\mathbf{q}}(t) \quad (\text{B.1})$$

where Ω is the area of the film. The inverse transformation is given by

$$\xi(\mathbf{q}, t) = \Omega^{-1/2} \int d^2 \mathbf{r} e^{-i\mathbf{q} \cdot \mathbf{r}} \xi(\mathbf{r}, t) \quad (\text{B.2})$$

This leads to

$$\begin{aligned} \langle \xi_{\mathbf{q}}(t) \xi_{\mathbf{q}'}(t') \rangle &= \Omega^{-1} \int d^2 \mathbf{r} d^2 \mathbf{r}' e^{-i(\mathbf{q} \cdot \mathbf{r} + \mathbf{q}' \cdot \mathbf{r}')} \\ &\quad \left| \langle \xi(\mathbf{r}, t) \xi(\mathbf{r}', t') \rangle \right| \end{aligned} \quad (\text{B.3})$$

$$= -2D\Omega^{-1} \delta(t - t') \int d^2 \mathbf{r} d^2 \mathbf{r}' e^{-i(\mathbf{q} \cdot \mathbf{r} + \mathbf{q}' \cdot \mathbf{r}')} \nabla^2 \delta(\mathbf{r} - \mathbf{r}') \quad (\text{B.4})$$

$$= -2D\Omega^{-1} \delta(t - t') \int d^2 \mathbf{r} d^2 \mathbf{r}' \delta(\mathbf{r} - \mathbf{r}') \nabla^2 e^{-i(\mathbf{q} \cdot \mathbf{r} + \mathbf{q}' \cdot \mathbf{r}')} \quad (\text{B.5})$$

$$= 2D\mathbf{q}^2 \Omega^{-1} \delta(t - t') \int d^2 \mathbf{r}' e^{-i(\mathbf{q} + \mathbf{q}') \cdot \mathbf{r}'} \quad (\text{B.6})$$

$$= 2D\mathbf{q}^2 \delta_{\mathbf{q}, -\mathbf{q}'}^2 \delta(t - t') \quad (\text{B.7})$$

In arriving at eqs B.4 and B.6, use has been made of eq 8d and Green's theorem, respectively. To convert eq B.7 from an expression appropriate to a discrete k -space to a continuous one, we replace $\Sigma_{\mathbf{q}}$ by $\Omega/(2\pi)^2 \int d^2 \mathbf{q}$ and get

$$\langle \xi_{\mathbf{q}}(t) \xi_{\mathbf{q}'}(t') \rangle = 2D(2\pi)^{-2} \mathbf{q}^2 \delta^2(\mathbf{q} + \mathbf{q}') \delta(t - t') \quad (\text{B.7})$$

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Notes

The authors declare no competing financial interest.

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